

Mohaddeseh (Sahar) Hosseinzadeh

Computational Neuroscience researcher | Machine learning Engineer | Software developer

Check out my website: <https://synapticsahar.github.io> [SynapticSahar](#) [Linked In](#)

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Interests: Computational Neuroscience Machine Learning Biomedical Signal Processing Medical AI Product Development

EXPERIENCE

ML Engineer — Incoming Intern

Barcelona Supercomputing Center Jul 16 – Sep 30, 2026

- Joining the Digital Health Unit for a summer internship under the supervision of Paula Petrone, focusing on deep-learning-based detection and classification of cardiac events, including STEMI, NSTEMI, and other clinically relevant categories.

Researcher — Brain Metastability and MOP

CNS Lab, TCN Lab, UPF Sep 2025 – Present

- Core question:** Does the dynamic structure of brain state transitions shape individual differences in cognition and decision-making?
- Computing whole-brain metastability via vortex-space and clustering, measuring Shannon and path entropy per subject, and correlating with cognitive flexibility and behavioral performance.
- Modeling maximum occupancy principle and measuring its performance by comparing predicted transition probabilities to empirical ones
- Stack:** fMRI, HCP, Vortex Space, Kuramoto, Entropy, k-means, Ridge Regression, MATLAB

Researcher — Sharp Wave Ripple Detection from LFP

CMC Lab Apr 2025 – Feb 2026

- Built full SWR detection pipeline without manual threshold tuning: NMF-based spectral feature learning, Transformer-based modeling, and session-level evaluation
- Stack:** PyTorch, Transformers, NMF, scikit-learn, NumPy

ML Engineer — EEG Emotion Recognition

Imam Khomeini Hospital · Clinical Lab Feb – Aug 2025

- Built full EEG emotion recognition pipeline: bandpass filtering, ICA artifact removal, epoching on clinical EEG using MNE-Python. Trained EEGNet for 4-class emotion detection (valence/arousal quadrants) from frequency band power features (theta, alpha, beta, gamma). Deployed real-time inference: FastAPI backend + Streamlit frontend. Dockerized on AWS EC2.
- Stack:** MNE-Python, EEGNet, PyTorch, FastAPI, Streamlit, Docker, AWS EC2

ML Engineer — Deep Learning on EEG/EMG — Wearable BCI

Myant Inc., Canada Jan 2024 – Jan 2025

- Preprocessed EEG/EMG from textile wearables; built eye tracking and sleep staging models from a headband; developed RNN for EMG-based AR/VR movement steering using a glove dataset.

Full-Stack Software Development

Rahnema College Mar 2021 – Mar 2022

- Built and deployed a full-stack web app end-to-end.

PROJECTS

Deep Q-Network for the Beer Game

Bachelor's Thesis, 2023

- PyTorch · DQN · Multi-agent RL — extended DQN for multi-agent inventory control; tested under varying demand conditions.
Code: [Github](#)

EEG → Image Decoding via CLIP (THINGS-EEG2)

2025

- Transformer encoder aligns EEG to CLIP embeddings via contrastive loss (NT-Xent); retrieves top-3 image categories the brain saw. GPT-4o narrates the decoded perception.
- Stack:** CLIP, Contrastive Learning, Transformers, MNE-Python, HuggingFace, OpenAI API

Seizure Detection — Fine-Tuned Foundation Model

2025

- Fine-tuned PatchTST on CHB-MIT Scalp EEG (PhysioNet); benchmarked against clinical baselines. Deployed via FastAPI + Streamlit on AWS SageMaker.
- Stack:** PatchTST, HuggingFace, Clinical EEG, FastAPI, AWS SageMaker

Synaptic Plasticity Modelling in Mouse V1

Sep 2024 – Apr 2025

- Brian2 · STDP · Allen Institute Ophys Dataset — simulated reward-modulated STDP in SNNs; studied VIP/SST/excitatory interactions under rewarded vs. passive conditions.
Code: [Github](#)

Shannon Entropy for fMRI Task Prediction (GRU)

July 2024

- PyTorch · HCP · Transfer Learning — GRU with cross-subject transfer; entropy features outperformed raw BOLD across all runs.
Code: [Github](#) Cert: [Here](#) Talk: [Here](#)

3D U-Net for Brain Tumor Segmentation

Feb 2024

- PyTorch · BraTS2020 · Dice loss — 3D U-Net for volumetric MRI segmentation with augmentation and LR scheduling.
Code: [Github](#)

EDUCATION

M.Sc., Computational Neuroscience

Pompeu Fabra University Sep 2025 – Sep 2026

B.Sc., Industrial Engineering

Sharif University of Technology 2017 – 2023

- Teaching Assistant: Linear Algebra (40+ undergrads)

SKILLS

Languages & Scripting

Python, MATLAB, C/C++, R, Java, JavaScript, Shell

ML / Deep Learning

PyTorch, TensorFlow, Keras, scikit-learn, SciPy

CNNs, RNNs/GRU, Transformers, RL, SNNs, DQN

Signal Processing

LFP, EEG, EMG, fMRI

LLMs & GenAI

HuggingFace, LangChain, RAG, OpenAI API, PEFT/LoRA, Prompt Engineering

Neuro Tools

Allen SDK, NEST, Brian2, PyNN, NEURON

Cloud / HPC

AWS (EC2, SageMaker, Lambda, Lightsail), Linux

Software Practices

Clean Code, DRY, KISS, YAGNI, SOLID, OOP, Design Patterns

Git, REST APIs, CI/CD, Docker, Agile/Scrum

INTERNSHIPS

Computational Neuroscience

Neuromatch Academy July 2024

- Certificate: [Here](#)

REFERENCES

Paula Petrone

[Barcelona supercomputing center](#)

Dr. Ruben Moreno-Bote

[TCN Lab, UPF](#)

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[CNS Lab, UPF](#)

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